A

PROJECT REPORT ON

" STATISTICAL ANALYSIS OF FACTORS INFLUENCING THE SUBJECT CHOICE AMONG GRADUATE-LEVEL STUDENTS "

Submitted in the partial fulfillment of degree of

Bachelor of science (BSc)

Submitted By,

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B.S.c-III year

Guided By,

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Department of Statistics
Shri Shivaji Science College, Amravati

2024-2025

DECLARATION

To

Principal

Shri Shivaji Science College,

Amravati.

Respected Sir,

I the undersigned, hereby declare that the project work entitled **"Statistical analysis of factors influencing the subject choice among graduate-level students"** submitted to **'Department of Statistics'** is my group work . This is our original work and has not been submitted anywhere for any degree/diploma. The system presented herein has not been duplicated from any other source.

I understand that any such copying in liable to be punished in any way the department authority may deem fit.

Thanking You.

Place: Amravati

Date:

Yours Sincerely,

Ritisha Purushottam Atone

BSc-III (Sem-VI)



Shri Shivaji Education Society, Amravati's

SHRI SHIVAJI SCIENCE COLLEGE,

Shivaji Nagar, Amaravati (M.S) 444603

NAAC: A+ Grade (CGPA 3.42), PM UShA: Component-3 Grant (Strengthen College), UGC: CPE, NIRF - 2024: Rank - 99, DBT: STAR College (Strengthening Component), DST: FIST, ISO: 9001- 2015, SGBAU: Lead College, Career Katta: Centre of Excellence

Certificate

This is to certify that Ms. **Ritisha Purushottam Atone** Class B.Sc. semester VI, has successfully completed the project titled "entitled **"Statistical analysis of factors influencing the subject choice among graduate-level**"under the guidance of **Dr. Shaziya Farah Mam** in Partial Fulfillment of the requirements for the Degree of Bachelor of Science (B.Sc.) under the curriculum of Sant Gadge Baba Amravati University, Amravati during the academic year 2024-2025.

Date: 21/04/2025
Place: Amravati

Dr. Shaziya Farah Mam
(Project Supervisor)

Dr. Ujwala Junghare Mam
(Head In Charge)

ACKNOWLEDGMENT

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CHAPTER : 1

INTRODUCTION

1.1 ABSTRACT

The choice of academic subjects at the graduate level is a critical decision that significantly influences students' career trajectories, personal growth, and contributions to their respective fields. This study aims to statistically analyze the various factors that affect subject selection among graduate-level students. The research utilizes quantitative methods to examine variables such as socio-economic status, academic performance, parental and peer influence, career aspirations, availability of resources, and personal interest.

Data were collected through structured questionnaires administered to a diverse sample of graduate students across multiple disciplines. Statistical tools, including chi-square tests, and correlation analysis, were employed to identify and measure the strength of associations between influencing factors and subject choice.

The findings indicate that while personal interest and career goals are the most significant factors, external elements such as family expectations and financial considerations also play notable roles. The study highlights the importance of a supportive academic environment and targeted career counseling in guiding students toward suitable academic paths.

This research contributes valuable insights for educators, policymakers, and academic advisors seeking to understand and support student decision-making processes. It also lays the groundwork for further investigations into improving academic and career planning strategies in higher education.

1.2 OBJECTIVES

1. To identify and categorize the key personal, academic, and socio-economic factors that influence subject choice among graduate-level students.

2. To examine the relationship between various influencing factors (e.g., interest, parental influence, job prospects, prior academic performance) and students' chosen fields of study.

3. To analyze the significance and strength of each factor using statistical tools such as correlation, chi-square tests, and regression analysis.

4. To develop a predictive model for subject choice based on the most influential variables identified.

5. To provide recommendations to academic advisors and educational institutions for improving student guidance and program offerings based on data-driven insights.

1.3 SCOPE OF STUDY

This study focuses on analyzing the various factors that influence the selection of academic subjects among graduate-level students. The scope is limited to:

- **Target Population:** Graduate-level students enrolled in programs across (B.Sc 1st ,2nd & 3rd year) of Shri Shivaji Science College ,Amravati.

- **Geographical Coverage:** The study will be confined to the urban and rural region students enrolled at Shri Shivaji Science College ,Amravati.

- **Data Collection Method:** A structured questionnaire is prepared an use to gather primary data from students, covering aspects such as personal interests, parental and peer influence, socio-economic background, academic performance, and career aspirations.

- **Time Frame:** The data is collected during the current academic session of 2024-2025.

- **Analytical Techniques:** Descriptive statistics, correlation analysis, and hypothesis testing will be applied to determine the significance and strength of relationships between independent variables and subject choice.

- **Limitations:** The study is limited to self-reported data and may not account for all possible external influences. Also, the findings may not be generalizable beyond the selected institutions or region.

1.4 SIGNIFICANCE OF STUDY

The selection of academic subjects at the graduate level plays a crucial role in shaping students' future career paths, personal development, and contributions to society. Understanding the factors that influence these decisions is essential for educators, policymakers, and academic institutions aiming to provide relevant guidance, curriculum design, and support systems.

This study is significant in several ways:

1. For Educational Institutions: The findings will help universities and colleges identify the key drivers behind students' subject choices, enabling them to develop tailored academic programs and support services that align with student needs and aspirations.

2. For Policymakers: By understanding the statistical relationships between various influencing factors (such as socio-economic background, career goals, parental influence, and academic performance), policymakers can design more effective educational policies and scholarship programs.

3. For Career Counselors and Educators: The insights gained can enhance the effectiveness of career counseling services by allowing counselors to consider data-driven factors that genuinely impact students' decisions.

4. For Future Researchers: This study contributes to the limited body of literature that statistically analyzes subject choice at the graduate level. It can serve as a foundation for future comparative or longitudinal studies across different regions or educational systems.

5. For Students: The study helps in raising awareness among current and prospective students about the various factors that may influence their academic decisions, allowing for more informed and conscious choices.

In essence, this research provides empirical evidence and actionable insights that can lead to better educational outcomes, equitable access to opportunities, and more strategic academic planning at the graduate level.

CHAPTER : 2

RESEARCH METHODOLOGY

2.1 Materials

Survey Instrument: A structured questionnaire created using Google Forms to gather data on factors influencing subject choice. The form included questions regarding demographic information, personal interests, career aspirations, peer influence, and institutional factors.

Statistical Software: Google Sheets or Excel for data organization and performing descriptive statistical analysis and testing.

2.2 Methods

Survey Design and Distribution: we created a questionnaire using Google for collecting data on factors influencing subject choice. The survey included questions about future scope , personal interests, career opyion , and peer influence.

The form was distributed to students across various departments at Shri Shivaji Science College, Amravati via email and departmental group . we selected 13 department from college ,our target sample size was 50 student from each department ,giving total sample size of 650.

Data Collection: Responses were automatically collected in Google Sheets, with each submission recorded for analysis. A total of 650 responses were collected using stratified sampling.

Data Cleaning: Incomplete or invalid responses were excluded to ensure data quality. Thus, resulting final sample of 174 students.

Descriptive Statistical Analysis: Frequencies and Percentages were calculated using Excel or Google Sheets to summarize the data and identify trends in the factors influencing subject choice**.**

Hypothesis Testing: Testing is performed, and formulae are applied to find the relation between various factors. Chi-Square test and Spearman's rank corelation formula are chosen for the study.

CHAPTER : 3

STATISTICAL ANALYSIS

Data Preparation : The data collected from Google Forms was exported to Excel Sheets for analysis.

After removing incomplete or invalid responses, the data was organized into categories for easy access and analysis, with each factor (e.g., subject preference , future scope) being assigned a separate column.

Descriptive Statistics : The primary analysis conducted was descriptive statistics, focusing on summarizing and organizing the data. The following statistical measures were used.

Frequencies : The number of times each factor or option was selected by the respondents. This helped in understanding which factors were most influential in the subject choice of students.

Percentages : The proportion of respondents who selected each factor. For example, calculating the percentage of students who chose "personal interest" as the primary reason for selecting their subject.

Data Processing: Using Excel or Google Sheets, functions such as COUNTIF were applied to calculate the frequency of responses for each question.

Percentage calculations were performed using basic formulae to provide a clear interpretation of the results.

Presentation of Results : The results were presented in the form of tables and charts (e.g., pie charts, bar graphs) to visually represent the distribution of responses for each influencing factor.

The findings were summarized to identify trends, such as the most common factors influencing subject choice, and the variation across different student groups (e.g., departments).

Testing of Hopothsis : Various Hypotheses are set to explore the objectives of this study. Chi-square test of independence and association of attributes are used to test the association between various factors like gender and students enrolment in the college, different influencing factors for the subject (students interest in the subject, availability of study material for the subject, , easy to understand and scoring subject in view of the student) and their locality status.

Rank corelation coefficient is used to find the relation between factors relating to choice of the college like (distance of the college, popularity of the college, better teaching at the college, suggested by seniors/friends/parents, various facilities provided by the college for students / Environment of the college) and the urban/rural area students.

3.1 : DATA REPRESENTATION

Table 1: Response Percent for Various Reasons of Choosing a Subject.

Reason for choosing the subject	Availability of study material	Easy to understand	Better teaching	Scoring subject	Intrest
Total responses	33	31	25	45	40
Percentage	18.97%	17.82%	14.33%	25.83%	22.99%

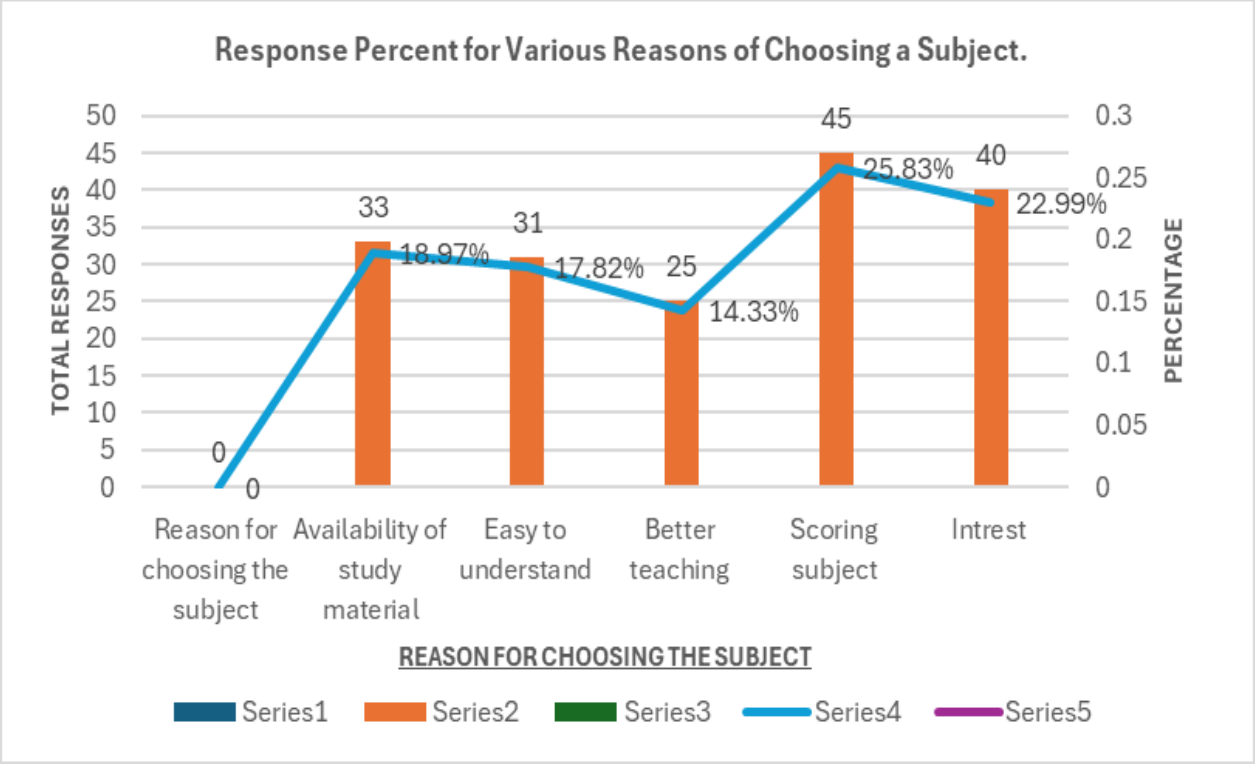


Figure No. 1

Result 1 :- According to the students' rankings, the primary reason for choosing a subject was that it is considered a scoring subject (32.1%). This was followed by the subject being easy to understand (26.4%), indicating a preference for clarity and simplicity. Availability of study material accounted for 23.5% of responses, showing the importance of accessible resources. The least influencing factor was better teaching (17.2%), suggesting that teaching quality was not a major deciding factor for most students.

Table 2:- Respondent Percentage Department wise

Group	Percentage
Botany	31.6 %
Chemistry	30.4 %
Zoology	15.5 %
Data analytics	29.8 %
Biotechnology	10.3 %
Microbiology	13.7 %
Computer science	64.9 %
Statistics	58.6 %
Physics	6.3 %
Mathematics	34.4 %
Environmental science	5.1 %

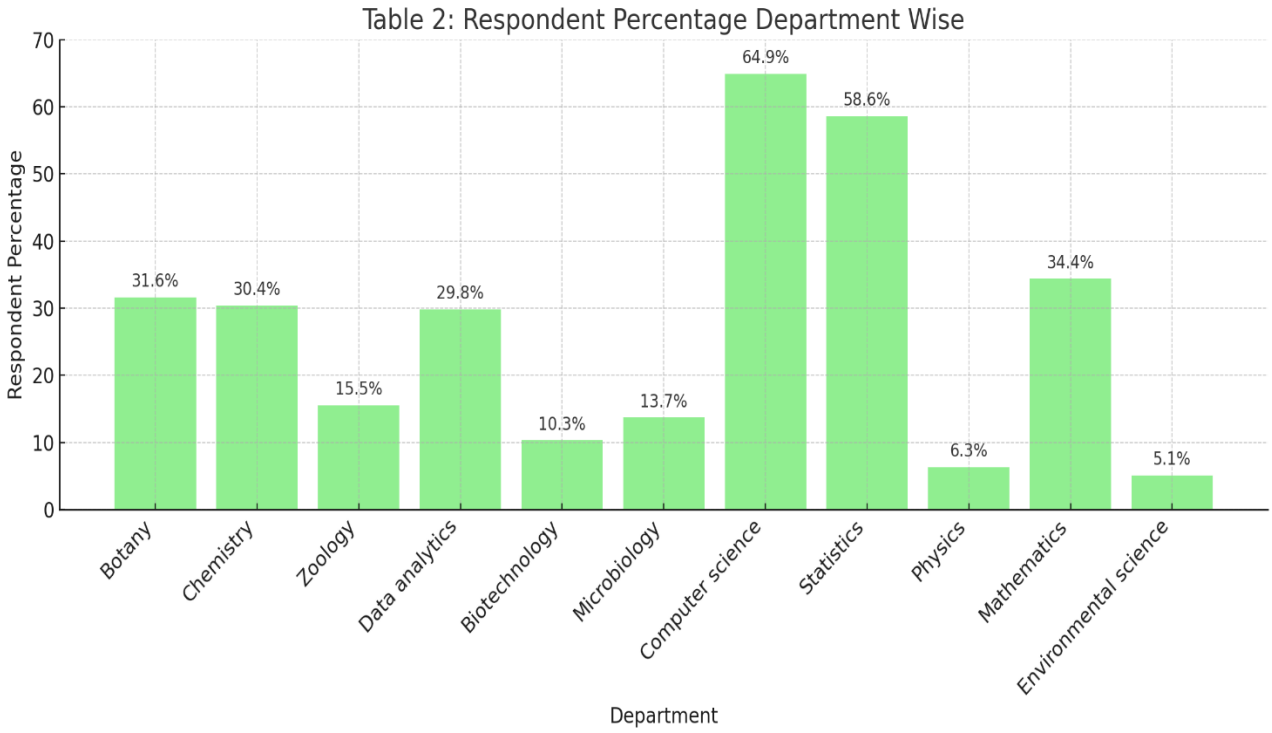


Figure No. 2

Result 2:-

The analysis of group-wise participation based on a total count of 174 reveals that Computer Science has the highest representation, with 64.9%, followed closely by Statistics at 58.6%. Mathematics also shows a strong presence, accounting for 34.5%. Among the natural sciences, Botany leads with 31.6%, followed closely by Chemistry at

30.4%, while Zoology trails behind at 15.5%. In the applied sciences, Data Analytics has a significant contribution of 29.9%, followed by Microbiology at 13.8% and Biotechnology at 10.3%. On the lower end, Physics accounts for only 6.3%, and Environmental Science has the least representation at 5.2%. These results highlight a strong inclination toward computational and data-oriented disciplines, while traditional and environmental sciences appear less represented in comparison.

Table 3:- Average of SSC & HSC Scores Across the Subjects Groups,

Subject Group	S.S.C.Average	H.S.C Average
Botany	53.1	47.8
Chemistry	60.9	54.3
Zoology	59.5	51.5
Data analytics	56.7	49.8
Biotechnology	48.5	42.9
Computer science	59.3	51.7
Microbiology	37.2	35.9
Statistic	59.5	51.7
Physics	64.2	47.2

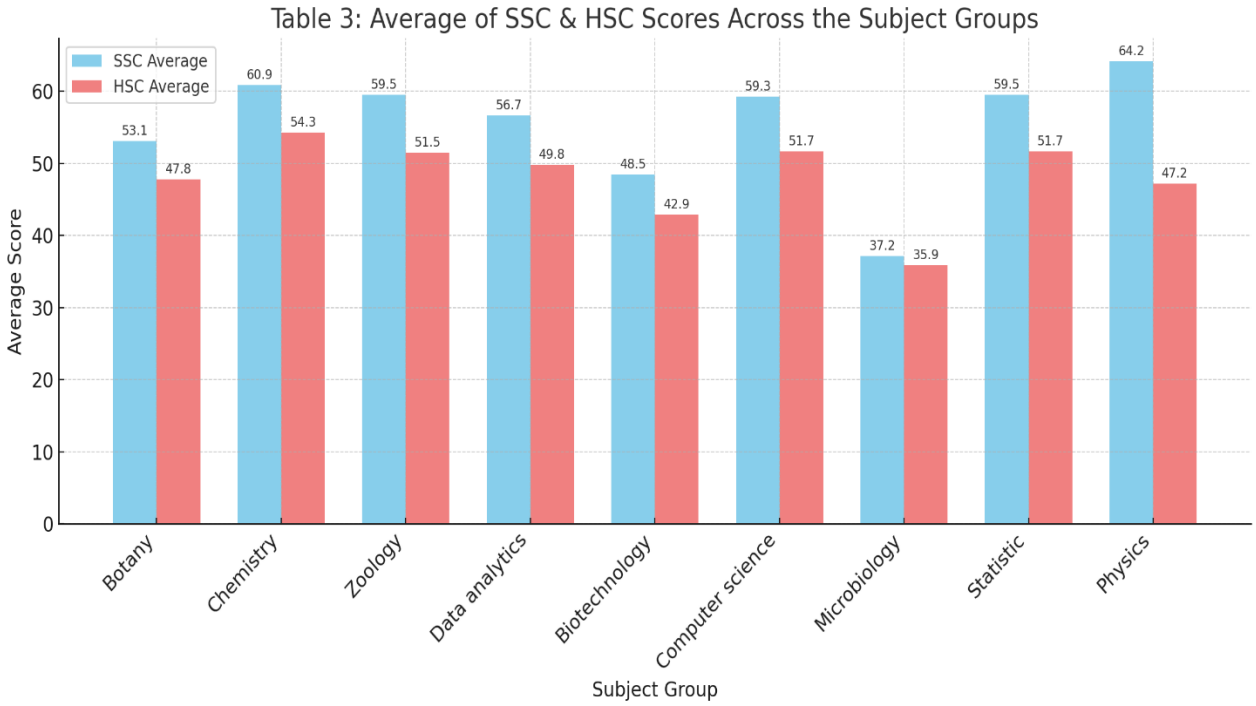


Figure No. 3

Result 3 :-

1) S.S.C averages are higher than H.S.C across all subjects
2) Physics shows the highest S.S.C score (64.2) but a significant drop in H.S.C (47.2).
3) Chemistry performs strongly in both exams.
4) Microbiology has the lowest scores.
5) Statistics remains consistent across both levels.
6) Tech subjects like Computer Science and Data Analytics show good overall performance.

Table 4:- Students Background Rural and Urban

Area	No. of Students	percentage
Ruler area	92	52.8 %
Urban area	82	47.1 %

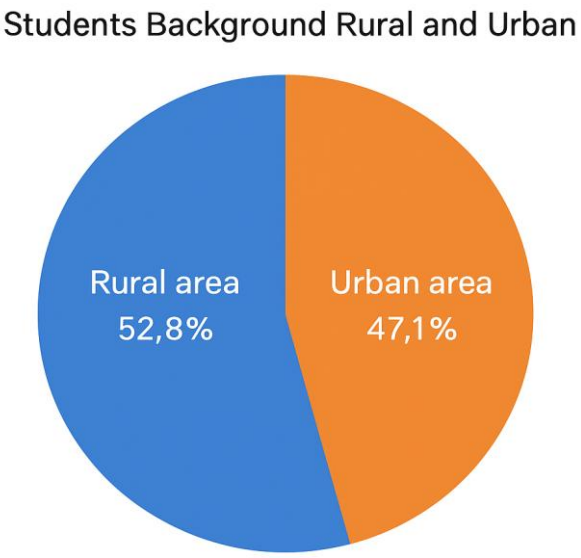


Figure No. 4

Result 4:-According to the survey data, a larger proportion of students belong to rural areas compared to urban areas. This indicates that rural students are actively participating in higher education, possibly due to improved access and awareness.

Table 5:-Reason of Choosing the College.

Reason	No. of student	Percentage
Distance of the college	13	7.4 %
Other Facilities/ Environment	68	39 %
Popularity of College	61	35 %
Suggested by someone	46	26.4 %

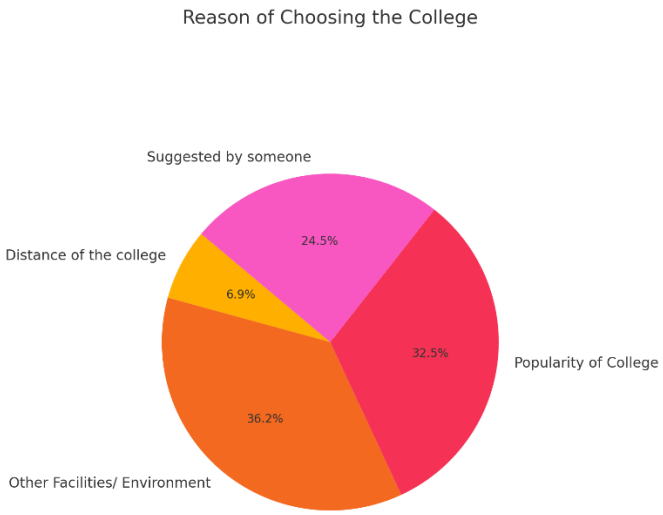


Figure NO:5

Result 5 :- Most students (39%) chose the college because of good facilities and environment. The second main reason was the popularity of the college (35%). Some students (26.4%) said they chose it because it was suggested by someone(Parents/seniors/friends) . Only a few students (13%) chose the college based on distance. This shows that students mostly care about how good the college is rather than how close it is.

Table 6:- Preference Between College and Academic Group

Preference	No. of Student	Percentage
College	130	74.7 %
Group	43	24.7 %

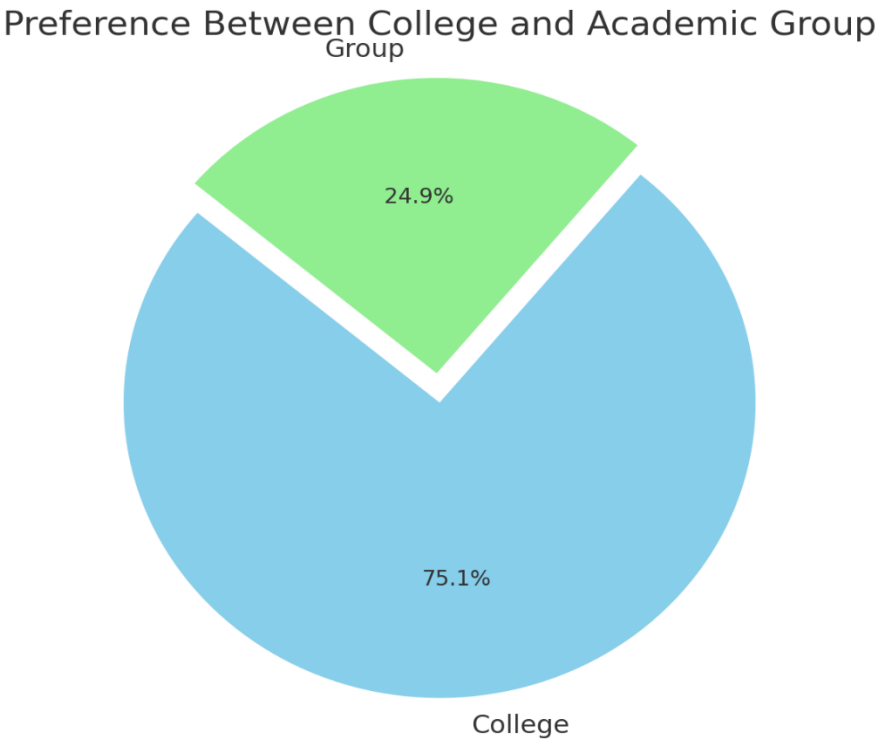


Figure No. 6

Result 6 :- The majority of students (74.7%) prefer selecting their college over a specific group or stream, which only 24.7% of students prioritize. This indicates that the reputation, facilities, or location of the college plays a more significant role in student decision-making compared to the academic group or stream offered.

Table 7:- Future Career Preferences of Students

Future Scope	No.of Student	Percentage
In government	66	37.9%
In private	45	25.8%
Business	33	18.9%
Research	28	16%

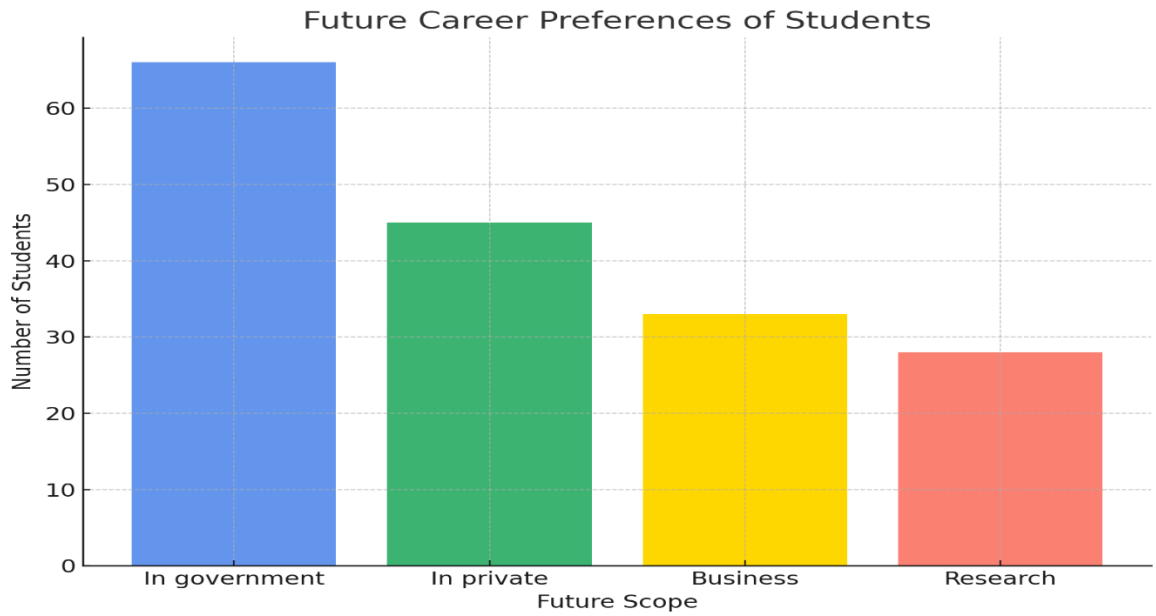


Figure No. 7

Result 7:- The majority of students (37.9%) expressed a preference for careers in the government sector, followed by 25.8% aiming for the private sector. 18.9% of students plan to go into business, while 16.0% are interested in pursuing research.

Table 8:- First preference of subject among student

Subject	No. of Student	Percentage
Botony	22	12.6%
Zoology	11	6.3%
Biotechnology	5	2.8%
Chemistry	8	4.5%
Computer science	48	27.5%
Computer application	23	13.2%
Data analytics	18	10.3%
Environmental science	5	2.8%
Statistics	14	8%
Microbiology	7	4%
Mathematics	11	6.3%
Physics	6	3.4%

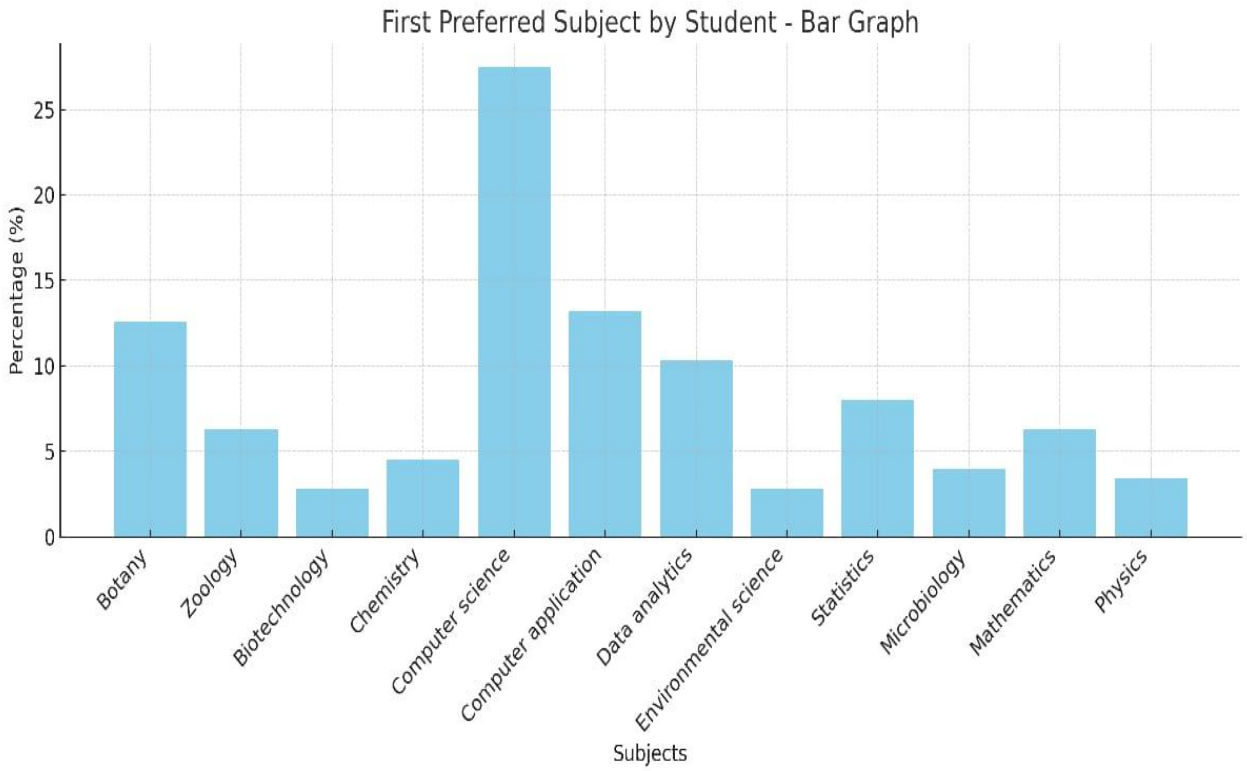


Figure No. 8

Result 8 :- The most preferred subject among students is Computer Science, chosen by 27.5% of students. This is followed by Computer Application (13.2%) and Botany (12.6%). Other notable preferences include Data Analytics (10.3%) and Statistics (8%). Subjects like Biotechnology, Environmental Science, and Physics were less preferred, each chosen by fewer than 5% of students. This shows a strong interest in technology and computer-related fields among student.

Table 9: Second preference subject among student

subject	No. of student	Percentage
Botony	16	9.1%
Zoology	28	16%
Chemistry	12	6.8%
Environmental Science	6	3.4%
Mathematics	23	13.2%
Computer science	12	6.8%
Computer application	2	1.1%
Statistics	71	40.8%

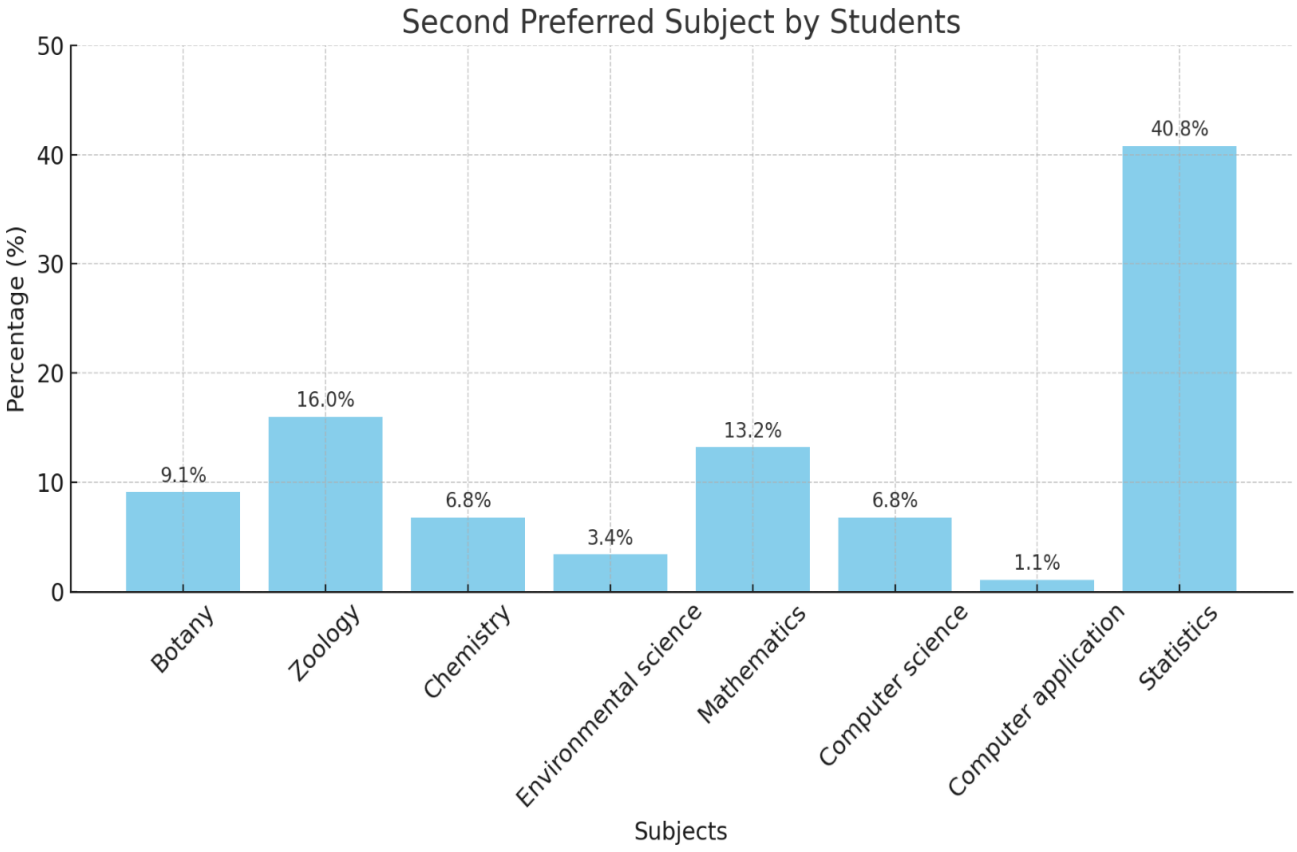


Figure No. 9

Result 9 :- This data shows that Statistics is the most popular second preferred subject among students, with 40.8% choosing it. This indicates a strong interest in analytical and data-related fields. Zoology (16%) and Mathematics (13.2%) are also notable second choices, reflecting continued interest in life sciences and numerical subjects. Botany (9.1%), Chemistry and Computer Science (6.8% each) show moderate preference. Subjects like Environmental Science (3.4%) and Computer Application (1.1%) have the least preference as a second choice.

3.3 : HYPOTHESIS TESTING

Chi-square test of independence and association of attributes is used to find relation between various factors.

Ho: Students taking admission to Shri Shivaji science college are independent of Gender and area of the students.

	Rural	Urban	Total
Male	17	12	29
Female	75	70	145
Total	95	82	174

Chi-sqr cal by exel formula = 0.497

p value = 0.05

Since chi sq cal > p value: Ho is accepted

Chi square calculation by formula:

oi	Ei	(Oi-Ei)^2	(Oi-Ei)^2/Eı
17	15.33	2.7889	0.181924331
75	76.66	2.7556	0.035945734
12	13.66	2.7556	0.201727672
70	68.33	2.7889	0.040815162
			0.4604129

chi square tab (1,1) at 5% l.s = 3.841, Ho: accepted

Result: Student admitted to Shri Shivaji science college are independent of gender and area

Ho : Students' locality Status and influencing factors for the subject are independent.

Student	Facilities & environment	Distance of College	popularity of College	Suggested by Someone	Total
Rural	44	6	23	27	100
Urban	32	15	14	13	74
Total	76	21	37	40	174

Calculation Table

Oi	43.6782	12.069	12.069	12.069
Ei	32.3218	8.93103	15.7356	17.0115

Chi square cal : 0.027231

p value = 0.05

0.03 < p-value (0.05)

Ho : Rejected

Result: There is a strong association between locality status (Rural/Urban) of the students and various factors influencing choice of the college.

CHAPTER 4 :

RESULT AND DISCUSSION

This study analyzed the factors influencing subject choice among graduate-level students at Shri Shivaji Science College, Amravati. Based on data from 650 students, it was found that *scoring potential, ease of understanding, and **availability of study material* were the top reasons for subject selection, while *teaching quality* had less impact.

Computer Science was the most preferred subject, and *Statistics* was the leading second choice, highlighting interest in technology and data-related fields. Most students performed better in *S.S.C.* than *H.S.C., and over half were from **rural areas*, reflecting broader access to higher education.

Statistical analysis also indicated that *gender and location of students are largely independent* of their subject choices, suggesting that academic interest and career goals are more influential.

Students mainly chose colleges based on *reputation and environment, and the **government sector* was the most preferred career path.

Overall, subject choices were shaped by a mix of *personal interest, practical benefits, and career aspirations*, offering insights that can help institutions provide better academic support and student guidance.

CONCLUSION

This study aimed to identify and analyse the primary factors influencing the subject choice among graduate-level students through statistical methods. The results indicate that personal interest and career opportunities are the most significant factors in decision-making. Additionally, academic performance, peer and family influence, and institutional support were also found to play key roles.

Demographic factors such as gender, socioeconomic status, and geographic location showed some influence, though their impact was less pronounced compared to the more direct factors like interest and career goals.

These findings highlight the need for educational institutions to understand these diverse influences and provide better guidance that aligns with students' individual goals and strengths. Tailoring support systems to students' aspirations could help in making more informed, personalized, and fulfilling academic decisions.

REFERENCE

1. Brown, P., & Hesketh, A. (2004). The Mismanagement of Talent: Employability and Jobs in the Knowledge Economy. Oxford University Press.

2. Trusty, J. (2002). Effects of high school course-taking and other variables on choice of science and mathematics college majors. Journal of Counselling & Development, 80(4), 464–474.

3. Iloh, C. (2018). Toward a new model of college “choice” for a twenty-first-century context. Harvard Educational Review, 88(2), 227–256.

4. Parker, P., Schoon, I., & Tsai, Y. M. (2011). Aspirations and reality: The role of social class and gender in university entry and subject choice in the UK. British Educational Research Journal, 37(3), 507–529.